

Project 4: Password Cracking

- Due Feb 22 by 11:59pm
- Points 100
- Submitting an external tool
- Available after Feb 10 at 12am

- **Do not use virtual machines (VMs) to complete this project— use your native computer for password cracking.**

- **NUwave will block some of the resources below— please use a different network if you are having trouble connecting.**

Overview: Description & Deliverables

In this project, you will gain hands-on experience with cracking passwords. **Start immediately**, as the necessary computations may take many days to complete. Although collaboration is not allowed on cracking, you may discuss general cracking strategies.

To Receive Full Credit, Submit:

1. **project4/cracked.txt** – A file containing the usernames and cracked passwords of your assigned users.
2. **project4/email.txt** – A file containing your Northeastern email (*not including .shadow*) in plaintext.

Background

Linux systems store hashed passwords in crypt format in the `/etc/shadow` file. In fact, if you have `sudo` access, you can view this file on your own system (*do not attempt this on Khoury College Linux machines!!*).

For this assignment, each student retrieves their unique file (`[username]@northeastern.edu.shadow`) from our [GitHub repository](https://github.com/cy2550repo/cy2550/tree/a06c0aff1ab94454c7b81deaf9e72a0cf5cbbc84/Project4)  (<https://github.com/cy2550repo/cy2550/tree/a06c0aff1ab94454c7b81deaf9e72a0cf5cbbc84/Project4>) of "leaked" shadow files (*please DM Justin ASAP if you do not see your email in this repository!!*). Each file contains 50 usernames and passwords to be cracked. **This requires at least several days of computing to crack all passwords. Start early!**

Password Cracking

Cracking Tools

You will be running either [John the Ripper](https://www.openwall.com/john/doc/) or [Hashcat](https://hashcat.net/wiki/doku.php?id=hashcat) to crack your passwords. We generally recommend Hashcat, but it does have a steeper learning curve so you may use John the Ripper if you prefer.

Installation Instructions

For Windows:

- **Hashcat:** Download the binaries from [Hashcat](https://hashcat.net/hashcat/) and extract them with [7-Zip](https://www.7-zip.org/). Run from PowerShell, within the `C:\hashcat` directory, with `.\hashcat64.exe [path_to_your_shadow_file]`.
- **John the Ripper:** Download the 7z, 64-bit binaries from [OpenWall](https://www.openwall.com/john/) and extract them with [7-Zip](https://www.7-zip.org/). Run from PowerShell, within the `C:\john-the-ripper-jumbo\run` directory, with `.\john.exe [path_to_your_shadow_file]`.

For Mac:

1. Install Homebrew: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
2. Update Homebrew: `brew update`
3. Install your preferred tool: `brew install john-jumbo` or `brew install hashcat`
4. Launch your preferred tool for the first time by running: `john [path_to_your_shadow_file]` or `hashcat [path_to_your_shadow_file]`

Cracking Approaches

1. **The easy passwords** (~ $\frac{1}{3}$ of the list): Can be cracked quickly (within minutes) by using a common wordlist and simple rules.
2. **The medium passwords** (~ $\frac{1}{3}$ of the list): Can be cracked within 24 hours by using a wordlist and built-in permutation rules or attack modes.
3. **The hard passwords** (~ $\frac{1}{3}$ of the list): Can be cracked in several days by using advanced techniques, such as custom permutation rules.
4. **The elites** (~3 passwords): Can only be brute-forced. This takes *many* days.

The Leaderboard— An Incentive to Start Early!

We have enabled Gradescope's leaderboard for this assignment. As you all submit, it will populate with the ten students who have cracked the most passwords. If you, at any point during the assignment, reach the top ten on the leaderboard, please DM a TA a screenshot of your name on the leaderboard and we will give you a hint on the project. The hint will be the same for all students: it is a link to the specific wordlist that Justin used to generate your shadow files. You can then use this wordlist to crack the remaining passwords more easily. A couple of notes:

- You must have cracked at least 5 passwords to be eligible for this hints (i.e., you cannot just be the first to submit with 0 passwords, you must actually start the assignment early)
- All students will be asked to provide their name for the leaderboard when submitting; you are free to use a pseudonym if you prefer, but please make sure we can verify it is you on the leaderboard if you send us a screenshot
- Only the top ten students will be shown on the leaderboard. Please do not feel bad if this is not you. The assignment can be completed without the wordlist hint, and your place in the class will not be publicly shown unless you are one of those top ten students.

Tips

1. When we say start early, we mean it. While this project is relatively hands-off it is impossible to complete last minute because the program itself (Hashcat or John the Ripper) needs to be run long enough to actually crack the passwords.
2. All students have different passwords. Please do not share cracked passwords with other students. Instead, discuss general cracking strategies.
3. If you have a GPU available, use it for faster cracking. If you do not, that is nothing to worry about.
4. Experiment! But also give your experiments time to run. Once you reach the harder passwords, the general rule of thumb is to run your command and if you do not have any additional passwords cracked within 24 hours, try something else.
5. The passwords are based on what real users might set as their password. Before you try writing any custom rules, think about what these passwords could possibly be.

Submission Instructions

1. Clone your cy2550 repository on your native computer if you have not done this already.
2. Create a directory: `~/cy2550/project4`.
3. Copy `cracked.txt` and `email.txt` into `~/cy2550/project4`.
 1. Your `cracked.txt` file must list each usernames and cracked password, separated by a colon, one per line:

```
1. alice:crackedpassword
   bob:crackedpassword
   charlie:crackedpassword
```
4. Push your repository to GitHub.

5. Submit your repository to Gradescope.

1. *Check the auto-graded output!!! Make sure it is what you expect!!! Fix it if not!!!*

Grading Breakdown

- 100 points for cracking **46** passwords
- **Extra credit** for passwords 47-50

Autograder Results

Results**Code****Leaderboard**

No match found for streplon
No match found for boosherha
No match found for fostrie
No match found for sparkbear
Found 46/50 cracked passwords!

Project 4: Password Cracking

 Graded

Select each question to review feedback and grading details.

Student

Hang Hang

Total Points